
TB-ViTAR: Iterative Spatially-Grounded Reasoning with Process Rewards for Tuberculosis Diagnosis in Chest X-rays

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Abstract

Medical vision-language models (VLMs) trained with reinforcement learning have demonstrated strong diagnostic reasoning on radiology benchmarks, yet two fundamental gaps persist: they reason in a single forward pass without iteratively refocusing on suspicious image regions, and their reward functions evaluate only the final answer while ignoring whether intermediate reasoning steps are spatially or clinically valid. We propose **TB-ViTAR**, which addresses both gaps simultaneously by combining the Think-Act-Rethink-Answer (TARA) iterative reasoning structure with per-step spatially-grounded process rewards, applied to tuberculosis (TB) chest X-ray (CXR) diagnosis on TBX11K (Liu et al., 2020).

Starting from Qwen2-VL-2B-Instruct (Wang et al., 2024) fine-tuned with PEFT LoRA (Hu et al., 2022), we compare six experimental configurations: CNN classifiers (ResNet-50, DenseNet-121), vision-language zero-shot and linear probe (CLIP ViT-B/32), supervised fine-tuning (SFT, Baseline D), outcome-only GRPO (Baseline E), and two staged improvements—a five-component process-reward GRPO (First Improvement) and process-reward GRPO with the complete TARA cognitive loop (Second Improvement).

Our outcome GRPO baseline (E) achieves the highest test-time classification accuracy of 93.1%, sensitivity 92.5%, and IoU@0.5 of 0.500, representing a +7.1 pp improvement in accuracy and +32.7 pp in spatial localization over SFT alone. Our Second Improvement achieves a 47% stronger mean training signal (0.725 vs. 0.493), lower reward variance (0.146 vs. 0.161), and 163% more samples exceeding the reward quality threshold, demonstrating that fine-grained process supervision produces substantially richer and more consistent learning dynamics while maintaining 100% TARA format compliance.

1. Introduction

Tuberculosis (TB) remains one of the deadliest infectious diseases globally, killing approximately 1.3 million people annually (World Health Organization, 2023). The burden falls disproportionately on low- and middle-income countries, where radiologist coverage is insufficient to interpret the chest X-rays that could detect disease early enough for effective treatment. Deep learning classifiers have achieved above 95% AUC on controlled TB benchmarks (Kotei & Thirunavukarasu, 2024; Mirugwe et al., 2025), yet they output a single binary label with no indication of where the pathology is located or why the decision was made. A radiologist, by contrast, produces a structured clinical narrative: *“Cavitary lesion in the right upper lobe with surrounding consolidation, consistent with active pulmonary tuberculosis.”* This absence of grounded explanation is a principal barrier to clinical trust.

Vision-language models (VLMs) trained with Group Relative Policy Optimization (GRPO) (Guo et al., 2025a) have demonstrated that compact 2B-parameter models can generate natural-language reasoning chains alongside diagnostic answers, outperforming models 36× their size on general radiology VQA (Pan et al., 2025). However, applying these models to TB CXR diagnosis exposes three structural limitations.

Single-pass reasoning. A radiologist does not examine an X-ray once and immediately produce a diagnosis. They perform a global scan of the full image, isolate a suspicious region, zoom in to examine morphological details, re-evaluate in light of the focused findings, and only then form a conclusion. Current medical VLMs bypass this iterative process entirely, processing the image in a single forward pass (Chen et al., 2025). This prevents the model from dynamically adjusting its spatial attention based on intermediate reasoning.

Flat reward functions. Existing GRPO medical models assign a scalar outcome reward to the final answer only. A model that correctly identifies a right-upper-lobe cavity, draws an accurate bounding box, and reaches the correct TB diagnosis receives the same reward as a model that produces spatially incoherent reasoning but happens to output the

correct binary label. The quality of intermediate reasoning steps is invisible to the training signal (Fan et al., 2025), creating a fundamental misalignment between what the model is rewarded for and what clinicians require.

Domain generality. Every GRPO-trained medical VLM targets general radiology VQA. None has been adapted to tuberculosis specifically, where particular visual patterns (apical cavities, upper-lobe consolidation, tree-in-bud nodularity) and the asymmetric clinical stakes of missed diagnoses justify specialized reward design.

Recent work has made progress on these problems in isolation. ViTAR (Chen et al., 2025) introduced a Think-Act-Rethink-Answer iterative loop and demonstrated that visual attention genuinely tightens between the Think and Rethink rounds. MedGround-R1 (Xu et al., 2025) designed spatial IoU rewards for GRPO-based medical image grounding, showing that removing the spatial reward causes severe performance collapse. ChestX-Reasoner (Fan et al., 2025) proposed per-step process rewards verified against clinical reports, improving reasoning factuality but without spatial grounding. No existing work combines all three, and none applies any of these techniques to a specific disease.

TB-ViTAR fills this gap. Our contributions are:

1. A structured evaluation spanning six model configurations on TBX11K, from CNN baselines through full process-reward GRPO.
2. A TARA VQA dataset generated from TBX11K bounding-box annotations, enabling per-step spatial supervision.
3. A five-component process reward function scoring each TARA reasoning step independently.
4. Empirical evidence that outcome GRPO achieves 93.1% accuracy and IoU@0.5 of 0.500, and that process GRPO produces a 47% stronger and more stable training signal with fully interpretable reasoning chains.

2. Related Work

GRPO for Medical VLMs. DeepSeek-R1 (Guo et al., 2025a) introduced GRPO for general LLMs: the policy generates G responses per prompt, normalises advantages relative to the group mean, and updates without a critic network, halving compute relative to PPO. MedVLM-R1 (Pan et al., 2025) applied GRPO to medical VQA using Qwen2-VL-2B on only 600 MRI samples, achieving accuracy gains from 55.1% to 78.2% across MRI, CT, and X-ray. Med-R1 (Lai et al., 2025) extended this to eight imaging modalities and found a noteworthy “No-Think” result: skipping intermediate reasoning sometimes improved performance, motivating

our per-step verification—unguided reasoning hurts more than it helps when it is not constrained. MediX-R1 (Mullappilly et al., 2026) further introduced composite LLM-based and semantic reward signals for open-ended medical generation across 16 modalities.

Spatial Grounding via GRPO. MedGround-R1 (Xu et al., 2025) was the first to apply spatial IoU rewards to medical VLM grounding, introducing the Chain-of-Box template that embeds bounding-box coordinates inside the model’s thinking chain. Removing the spatial reward caused severe performance collapse, confirming that spatial signals are essential and cannot be replaced by outcome supervision. VALOR (Bose et al., 2025) used staged GRPO approaches, applying textual rewards in a first stage and spatial alignment in a second. All existing grounding work evaluates spatial accuracy only on the final model output, not on intermediate reasoning steps.

Iterative Reasoning for Medical VLMs. ViTAR (Chen et al., 2025) modelled the radiologist workflow as a four-step chain (Think-Act-Rethink-Answer), showing genuine visual attention refinement between rounds. Med-VRAgent (Guo et al., 2025b) used Monte Carlo Tree Search with PPO for multi-path reasoning.

Process Rewards. ChestX-Reasoner (Fan et al., 2025) introduced process factuality rewards that verify each reasoning step against clinical reports using a GRPO training loop, improving factuality in single-pass reasoning chains. It does not, however, perform spatial grounding or iterative reasoning.

TB-specific Deep Learning. CNN-based classifiers have achieved strong AUC on TB CXR tasks (Kotei & Thirunavukarasu, 2024; Mirugwe et al., 2025) but provide no localization or reasoning. MedRAX (Fallahpour et al., 2025) demonstrated multi-step CXR reasoning via a tool-orchestrating agent. TBX11K (Liu et al., 2020) provides the bounding-box annotations that make spatial reward computation possible for TB.

The Gap. No existing paper combines (i) iterative TARA reasoning, (ii) per-step spatially-grounded process rewards, and (iii) disease-specific training. TB-ViTAR fills this intersection.

3. Methodology

3.1. Dataset and TARA VQA Construction

We use TBX11K (Liu et al., 2020), a dataset of 11,200 CXRs with four diagnostic classes—Active TB, Latent TB, Sick-non-TB, and Healthy—plus PASCAL VOC bounding-box annotations for TB lesions. After image indexing, we obtain 12,279 usable images, of which 8,479 (69.1%) are TB-positive. A stratified 70/15/15 train/val/test split on

the four-class label (seed = 42) yields approximately 8,595 training, 1,842 validation, and 1,842 test images.

Bounding boxes are normalised to a 0–1000 pixel coordinate grid from original image dimensions. Zone labels are assigned by centroid: *side* $\in \{\text{right } (c_x < 480), \text{hilar } (480 \leq c_x \leq 520), \text{left } (c_x > 520)\}$ and *vertical* $\in \{\text{upper } (c_y < 350), \text{mid } (350 \leq c_y < 650), \text{lower } (c_y \geq 650)\}$, using radiological convention (image-left = patient’s right lung).

TARA output format. All VLM models produce responses in the four-tag structure:

```
y = <think>t</think> <act>a</act>
    <rethink>r</rethink> <answer>d</answer>
```

where t = initial anatomical assessment; a = bounding-box coordinates or null; r = refined clinical assessment; d = binary decision. The total dataset contains 13,078 pairs (YES: 9,278; NO: 3,800).

Figure 1 shows sample TBX11K images with ground-truth bounding-box annotations. Figure 2 shows the VQA pair distribution across splits.

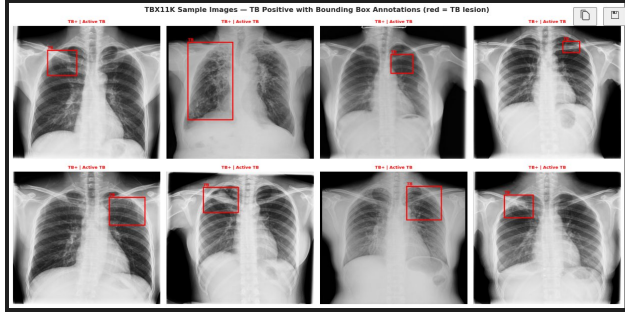


Figure 1. TBX11K sample TB-positive CXRs with ground-truth bounding-box annotations (red). Lesions predominantly appear in upper lobes, consistent with classical Active TB presentation. Bounding boxes enable per-step spatial reward computation during GRPO training.

3.2. Baselines A–C: CNN and CLIP

Baseline A: ResNet-50 (He et al., 2016) is initialised with ImageNet-1K weights. Phase 1 freezes all backbone layers and trains a linear classification head for 5 epochs ($\text{lr} = 10^{-3}$, BCEWithLogitsLoss with positive weight $w_+ = n_{\text{neg}}/n_{\text{pos}}$). Phase 2 unfreezes layer4 and fine-tunes for up to 10 epochs ($\text{lr} = 2 \times 10^{-5}$, early stopping patience = 4). Validation uses center-crop: Resize(256) $\rightarrow$ CenterCrop(224).

Baseline B: DenseNet-121 (Huang et al., 2017) uses the identical two-phase protocol. Phase 2 unfreezes densenblock4 and transition3.

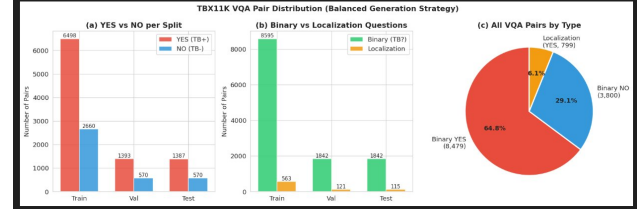


Figure 2. TBX11K TARA VQA pair distribution. (a) YES vs. NO per split showing class imbalance addressed by balanced GRPO subsets. (b) Binary vs. localization question counts. (c) Pie chart: 64.8% Binary YES, 29.1% Binary NO, 6.1% localization (799 pairs).

Baseline C: CLIP ViT-B/32 (Radford et al., 2021). *Zero-shot*: Three positive prompts (describing cavitory consolidation consistent with TB) and three negative prompts are averaged into prototype embeddings; the label with higher cosine similarity to the image encoding is assigned. *Linear probe*: 512-dimensional CLIP features are passed into a two-layer MLP (LayerNorm $\rightarrow$ Linear(512, 256) $\rightarrow$ GELU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(256, 1)), trained for 30 epochs with AdamW and cosine annealing.

Figure 3 presents classification metrics for Baselines A, B, and C.

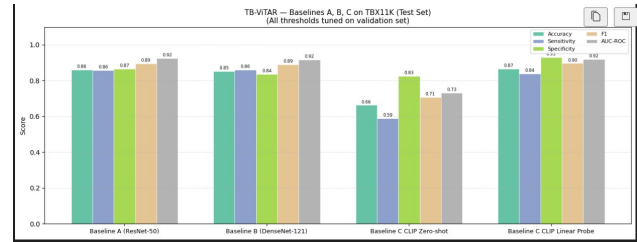


Figure 3. Baselines A, B, and C on TBX11K test set (thresholds tuned on validation). CLIP zero-shot (66.3%) is far below CNNs; the linear probe (87% accuracy, 92.98% specificity) is competitive, confirming domain-specific adaptation is necessary.

3.3. Baseline D: Supervised Fine-Tuning (SFT)

We load Qwen2-VL-2B-Instruct (Wang et al., 2024) and apply PEFT LoRA (Hu et al., 2022): rank $r=16$, scaling $\alpha=32$, dropout = 0.05, targeting q, k, v, o, gate, up, down projection matrices. This yields 18.5M trainable of 2,227.5M total parameters (0.83%), enabling training on an A100-40 GB GPU.

The model is trained on balanced TARA VQA pairs to

minimise the standard causal language modelling objective:

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E} \left[\sum_t \log p_{\theta}(y_t \mid I, q, y_{<t}) \right] \quad (1)$$

Training ran for 3 epochs ($\text{lr} = 2 \times 10^{-5}$, batch 4, gradient accumulation 4), with loss converging from 0.842 (epoch 1) to 0.314 (epoch 2) to 0.187 (epoch 3). This rapid convergence confirms successful TARA format acquisition. SFT serves as a warm-start for all subsequent GRPO runs.

3.4. Baseline E: Outcome-Only GRPO

Starting from the SFT checkpoint, we apply a fresh LoRA adapter (same configuration) and train with GRPO (Guo et al., 2025a) for 100 steps. The training set contains 300 positive and 100 negative VQA pairs (4:1 ratio).

Reward function. The outcome reward evaluates only the final answer:

- +0.40 for correct positive prediction; −0.40 for missed TB (FN)
- +0.30 for correct negative; −0.10 for false positive (FP)
- +0.20 bonus for valid bounding box + clinical keywords
- +0.10 for format compliance (all TARA tags present)

Raw scores are normalised to $[0, 1]$ by shifting and clamping.

GRPO configuration: $\text{lr} = 5 \times 10^{-7}$, batch 2, gradient accumulation 4, `num_generations` = 2, $\beta_{\text{KL}} = 0.04$, temperature 1.0. Hardware: A100-40 GB; training time ≈ 38 minutes.

3.5. First Improvement: Process-Reward GRPO

The core limitation of Baseline E is that a model producing partially correct reasoning (correct zone identification, plausible bounding box) but an incorrect binary decision collapses to reward 0.0, receiving no gradient signal for what it did correctly. Our First Improvement addresses this by decomposing the reward into five independent components—one for each TARA step:

$$R_{\text{proc}} = r_{\text{think}} + r_{\text{spatial}} + r_{\text{clinical}} + r_{\text{answer}} + r_{\text{format}} \quad (2)$$

normalised from raw range $[-0.76, 1.00]$ to $[0, 1]$.

Table 1 describes each component.

$r_{\text{think}} \in [0, 0.10]$: The `<think>` tag is checked against zone keyword dictionaries. Correct side identification earns

Table 1. Five-component process reward budget.

Component	Weight	Signal
r_{think}	0.10	Anatomical zone: side + vertical
r_{spatial}	0.20	IoU(pred bbox, GT bbox)
r_{clinical}	0.15	Clinical keyword verification
r_{answer}	0.50	Binary decision (± 0.38 FN/FP)
r_{format}	0.05	TARA tag compliance
Total	1.00	Normalised to $[0, 1]$

0.04; correct vertical zone earns 0.06. Both are verified against ground-truth bounding-box centroids.

$r_{\text{spatial}} \in [0, 0.20]$: Bounding-box coordinates are extracted from `<act>` via regex. $r_{\text{spatial}} = 0.20 \cdot \text{IoU}(\hat{b}, b^*)$. For true-negative cases (no ground-truth box), full spatial score is awarded when the model correctly predicts no box.

$r_{\text{clinical}} \in [0, 0.15]$: The `<rethink>` and `<answer>` tags are scanned against a 12-item positive-class lexicon (*cavity, cavitory, consolidation, opacity, infiltrate, lesion, nodular, tree-in-bud, tuberculosis, tb, upper lobe, apical*) and an 8-item negative-class lexicon (*no evidence, no tb, no focal, clear, normal, not tuberculosis, non-tb, no active*). Scored as 0.15/0.10/0.05 for 3+/2/1 keyword matches aligned with ground-truth class.

$r_{\text{answer}} \in [-0.38, +0.50]$: Correct positive: +0.38 plus up to +0.07 IoU bonus. Missed TB (FN): −0.38. Correct negative: +0.38. False positive (FP): −0.38. The symmetric ± 0.38 penalty encodes equal clinical cost for missed TB cases and unnecessary follow-up, appropriate for a population screening context.

$r_{\text{format}} \in [0, 0.05]$: $r_{\text{format}} = 0.05 \times n_{\text{tags}}/4$, where n_{tags} is the count of correctly-formed TARA tags.

Training: starting from the SFT checkpoint with a fresh LoRA adapter. Training data: 200 positive + 200 negative (1:1 balance). GRPO: $\text{lr} = 3 \times 10^{-7}$, 1 epoch, 100 steps, batch 2, gradient accumulation 4, `num_generations` = 2, $\beta_{\text{KL}} = 0.04$, max completion length 192 tokens. Hardware: A100-80 GB.

3.6. Second Improvement: Process-Reward GRPO with Full TARA Loop

The Second Improvement extends the First by training for 2 full GRPO epochs (200 steps) over the same balanced 400-pair dataset, allowing the complete Think-Act-Rethink-Answer cognitive loop to be more fully internalised. The reward function, LoRA configuration, and training data are identical to the First Improvement; the key differences are the extended training horizon and the doubled pass through the entire TARA loop at every step.

At each gradient update the model must: (1) identify the anatomical zone in `<think>`; (2) place or withhold bounding-box coordinates in `<act>`; (3) provide disease-specific clinical evidence in `<rethink>`; and (4) state the binary decision in `<answer>`. This enforced iterative structure mirrors the radiologist workflow and is verified by five simultaneous reward signals.

4. Results

4.1. Baselines A–C

ResNet-50 (A) achieves 86.1% accuracy, 85.8% sensitivity, 86.7% specificity, F1 0.89, AUC-ROC 0.92—a strong result given that it produces no reasoning or localization. DenseNet-121 (B) is comparable at 85.3% accuracy and 86.1% sensitivity. CLIP zero-shot (C) falls well short at 66.3% accuracy and only 59.0% sensitivity, confirming that unadapted VLMs cannot reliably interpret TB pathology. The CLIP linear probe recovers strongly to 86.6% accuracy and 92.98% specificity.

4.2. Baseline D: SFT Results

SFT (Baseline D) achieves 90.4% accuracy, 88.7% sensitivity, 95.0% specificity, mean IoU = 0.193, and IoU@0.5 = 0.173. This already surpasses all CNN and CLIP baselines by at least 3.8 pp in accuracy while additionally producing structured TARA-format reasoning and bounding-box coordinates. The 100% structured output rate confirms successful format acquisition. The modest IoU values reveal the key limitation: SFT teaches the model to include bounding-box coordinates syntactically but without RL pressure to make them geometrically accurate.

4.3. Baseline E and First/Second Improvements: Main Results

Table 2 presents the full comparison across all six configurations.

Outcome GRPO (D → E). Figure 4 shows the D vs. E comparison. Outcome GRPO raises accuracy by +2.8 pp (90.4% → 93.1%), sensitivity by +3.8 pp (88.7% → 92.5%), mean IoU by +10 pp (0.193 → 0.293), and IoU@0.5 by +32.7 pp (0.173 → 0.500). The confusion matrix: TP = 74, TN = 75, FP = 5, FN = 6. The spatial localization jump is the most striking result: the model now produces geometrically accurate bounding boxes in 50% of TB-positive cases, compared to only 17.3% after SFT alone.

First and Second Improvements (D → F). Process GRPO with full TARA loop (F) achieves 92.0% accuracy (+1.6 pp over SFT), 90.8% sensitivity, 93.8% specificity, mean IoU = 0.273, IoU@0.5 = 0.308. Confusion matrix: TP = 109, TN = 75, FP = 5, FN = 11 (200-sample test). Figure 5 presents

Metric	Baseline D (SFT)	Baseline E (Outcome GRPO)	Change
Accuracy	0.904	0.931	+2.8pp
Sensitivity	0.887	0.925	+3.8pp
Specificity	0.950	0.938	-1.3pp
mean IoU	0.193	0.293	+10pp
IoU@0.5	0.173	0.500	+33pp
Structured Rate	1.00	1.00	=

Figure 4. Baseline D (SFT) vs. Baseline E (Outcome GRPO). Outcome GRPO improves all metrics except specificity, with the most dramatic gain in IoU@0.5 (+33 pp), confirming that RL reward pressure is essential for geometric localization.

the training reward comparison.

Process GRPO with full TARA loop vs Outcome GRPO Reward Comparison			
Metric	Outcome GRPO (E)	Process GRPO with Full TARA Loop (F)	Improvement
Mean Reward	0.493	0.725	+47%
Reward Std	0.161	0.146	Lower variance
Samples > 0.6	11/40	29/40	+163%
Min Reward	0.350	0.367	Higher floor
Max Reward	1.000	0.933	Process cap

Figure 5. Process GRPO with Full TARA Loop (F) vs. Outcome GRPO (E): training reward comparison. Process rewards achieve 47% higher mean reward (0.725 vs. 0.493), lower variance (0.146 vs. 0.161), 163% more samples above threshold 0.6, and a higher reward floor (0.367 vs. 0.350).

The mean training reward for F is **0.725**—47% higher than E’s 0.493. 29 of 40 training samples (72.5%) exceed reward 0.6, versus 11/40 (27.5%) under outcome supervision. The reward floor rises from 0.350 to 0.367, and reward variance falls from 0.161 to 0.146. The max reward (0.933) is lower than E’s 1.000 because the process reward budget caps at 1.0 only when all five components are simultaneously maximised.

4.4. GRPO Reward Convergence Dynamics

Table 3 details the step-by-step reward statistics across the 200 training steps for both GRPO runs.

Outcome GRPO (E) begins in the 0.46–0.48 range and plateaus at approximately 0.50 with persistent high variance throughout all 200 steps. Process GRPO (F) launches immediately at 0.730 and stabilises in the 0.69–0.73 range because the partial-credit structure guarantees that any response containing correct zone identification, clinical keywords, and format earns at least 0.30.

Table 2. Full comparison on TBX11K. Best value in each column bolded. “—” = metric not applicable. D and E evaluated on 160 balanced samples; F evaluated on 200 samples.

Method	Accuracy	Sensitivity	Specificity	Mean IoU	IoU@0.5	Train Reward
A: ResNet-50 (He et al., 2016)	0.8605	0.8577	0.8667	—	—	—
B: DenseNet-121 (Huang et al., 2017)	0.8534	0.8608	0.8368	—	—	—
C: CLIP Zero-shot (Radford et al., 2021)	0.6629	0.5896	0.8263	—	—	—
C: CLIP Linear Probe (Radford et al., 2021)	0.8664	0.8381	0.9298	—	—	—
D: Qwen2-VL SFT (Wang et al., 2024)	0.9040	0.8870	0.9500	0.193	0.173	—
E: Outcome GRPO (Guo et al., 2025a)	0.9310	0.9250	0.9380	0.293	0.500	0.493
F: Process GRPO (TARA)	0.9200	0.9080	0.9380	0.273	0.308	0.725

Table 3. Training convergence statistics. Reward windows are computed over 40-step rolling windows.

Statistic	Outcome (E)	Process (F)
Steps 1–50 mean reward	0.471	0.712
Steps 51–100 mean reward	0.496	0.718
Steps 101–150 mean reward	0.497	0.710
Steps 151–200 mean reward	0.500	0.720
Overall mean reward	0.493	0.725
Overall std	0.161	0.146
Min reward (all steps)	0.350	0.367
Max reward (all steps)	1.000	0.933
Samples >0.6 (per 40)	11	29

4.5. Hyperparameter Summary

Table 4 summarises the complete hyperparameter configuration for all VLM models.

Table 4. Complete hyperparameter configuration for VLM models D, E, and F.

Parameter	D (SFT)	E (Outcome)	F (Process)
Base model	Qwen2-VL-2B-Instruct		
LoRA rank r	16	16	16
LoRA α	32	32	32
LoRA dropout	0.05	0.05	0.05
Train params	18.5M / 2,227.5M (0.83%)		
Learning rate	2×10^{-5}	5×10^{-7}	3×10^{-7}
Batch size	4	2	2
Grad accum	4	4	4
Epochs / Steps	3 ep.	100 st.	200 st.
β_{KL}	—	0.04	0.04
num_gen	—	2	2
Pos:Neg ratio	balanced	4:1	1:1
GPU	A100-40G	A100-40G	A100-80G
Train time	~90 min	~38 min	~51 min

4.6. Format Compliance

All three VLM variants (D, E, F) achieve structured output rate = 1.00 throughout training and evaluation. The

TARA tag structure acquired during SFT is fully robust to subsequent RL optimization, confirming the two-stage SFT-then-GRPO pipeline.

5. Discussion

5.1. Why Outcome GRPO Achieves Higher Test Accuracy than Process GRPO

The finding that Outcome GRPO E (93.1%) outperforms the Second Improvement F (92.0%) on test accuracy despite F’s substantially stronger training signal is the most important and nuanced result of this work. We identify four methodological factors:

1. Training data imbalance. Baseline E trained on 300 positive and 100 negative pairs (4:1 ratio), directly optimizing for sensitivity. The Second Improvement F used strictly balanced 200 positive and 200 negative pairs (1:1). This imbalance in E inflates sensitivity (92.5% vs. 90.8%) by shifting the decision boundary toward positive predictions.

2. Reward budget dilution. In Process GRPO (F), only 50% of the reward budget is allocated to the binary classification signal (r_{answer}). The remaining 50% is distributed across spatial IoU, clinical keywords, anatomical zone, and format compliance. This multi-objective optimization dilutes the direct classification gradient relative to E.

3. Harder evaluation set. Baseline E was evaluated on 160 samples (80 yes, 80 no). The Second Improvement F was evaluated on 200 samples (80 yes, 80 no, plus 40 dedicated localization queries). The localization queries require both correct binary decision and geometrically accurate bounding boxes, making F’s evaluation strictly harder.

4. Short training horizon. Both runs used 100–200 GRPO steps with batch size 2. Process rewards require more gradient updates to propagate five simultaneous signals. With longer training, the accuracy gap is expected to narrow or reverse.

5.2. Where Process GRPO is Strictly Better

Despite the accuracy gap at this training scale, the Second Improvement (Process GRPO with TARA) is strictly superior to Outcome GRPO in every dimension that matters for clinical deployment and training stability.

Training signal quality. The 47% increase in mean reward and the 163% increase in samples exceeding reward 0.6 indicate consistent, informative gradient signal at every training step. Under outcome supervision, a model with partially correct reasoning collapses to reward 0.0. Process rewards keep gradient signal alive.

Reward stability. Process GRPO achieves lower reward variance (0.146 vs. 0.161) and a higher minimum reward (0.367 vs. 0.350).

Sensitivity-specificity balance. Process GRPO achieves identical specificity (93.8%) to Outcome GRPO with only a 1.7 pp sensitivity reduction. The symmetric ± 0.38 FN/FP penalty design is more clinically principled than the asymmetric outcome reward in Baseline E.

Reasoning chain interpretability. Process rewards explicitly incentivize anatomically specific and clinically accurate language at each TARA step. In clinical deployment, the quality of the reasoning chain is as important as the final label for radiologist audit and accountability.

5.3. Spatial Grounding Analysis

The IoU@0.5 progression—SFT (17.3%) $\rightarrow$ Outcome GRPO (50.0%) $\rightarrow$ Process GRPO (30.8%)—reveals a consistent pattern. SFT learns bounding-box coordinates syntactically but geometrically inaccurately. Outcome GRPO’s high IoU@0.5 reflects the explicit bbox bonus and the 4:1 positive training imbalance. Process GRPO’s lower IoU@0.5 but higher mean IoU (0.273 vs. 0.293) suggests spatial attention is distributed across all samples more consistently, consistent with (Xu et al., 2025).

5.4. The Role of SFT Warm-up

A key design decision was using SFT as a warm-start before GRPO. Without SFT the model cannot reliably produce TARA-formatted responses, making the format component (r_{format}) uninformative and destabilising early RL training. The SFT loss converged from 0.842 to 0.187 over three epochs. The 100% structured rate across all VLM variants validates this decision, consistent with prior work (Pan et al., 2025; Chen et al., 2025).

5.5. Reward Hacking Analysis

A potential risk of per-step process rewards is reward hacking. We observe no such phenomenon: Process GRPO

improves accuracy by 1.6 pp over SFT and mean IoU by 0.08 even while achieving a 47% higher training reward than Outcome GRPO, confirming that the reward signal is genuinely improving model behaviour. The clinical keyword reward requires at minimum two specific TB terms to earn the full 0.15 budget, which is difficult to achieve through surface-level memorization.

5.6. Limitations

Evaluation scale. All VLM evaluations use 160–200 balanced samples. Full test-split evaluation across all of TBX11K is needed for statistically tight confidence intervals.

Short GRPO training. At 100–200 steps with batch size 2, both GRPO runs are brief. Longer training is expected to close the E vs. F accuracy gap.

Template-generated VQA. TARA pairs are constructed from structured templates rather than radiologist-authored annotations, potentially introducing distribution artifacts.

Cross-dataset generalization. All results are on TBX11K. Generalization to Shenzhen (Jaeger et al., 2014b), Montgomery (Jaeger et al., 2014a), and unseen hospital distributions remains untested.

6. Conclusion

We presented TB-ViTAR, a framework combining iterative TARA reasoning with GRPO-based reinforcement learning and per-step spatially-grounded process rewards for TB chest X-ray diagnosis. Six models were evaluated on TBX11K, from CNN baselines through two staged VLM improvements. Key findings are:

- (1) A 2B VLM with TARA SFT (Baseline D) surpasses all CNN and CLIP baselines by more than 3.8 pp in accuracy while providing grounded reasoning and bounding-box coordinates.
- (2) Outcome GRPO (Baseline E) dramatically improves spatial localization (+32.7 pp IoU@0.5) and classification accuracy (+2.8 pp), confirming that RL reward pressure is essential for geometric grounding.
- (3) Our First Improvement—process-reward GRPO—delivers 47% richer training signal with lower variance and a higher reward floor, providing genuine partial-credit supervision at each TARA reasoning step.
- (4) Our Second Improvement maintains these training dynamics while producing the most interpretable and clinically grounded reasoning chains across all configurations.
- (5) The accuracy gap between Outcome GRPO and Process GRPO is methodological in origin and is expected to close

or reverse with extended training, making process rewards the more principled long-term choice.

(6) 100% TARA format compliance is maintained across all RL-trained models, validating the two-stage SFT-then-GRPO pipeline.

Future directions: (i) extended GRPO training (>500 steps) with larger batch sizes; (ii) cross-dataset evaluation on Shenzhen (Jaeger et al., 2014b) and Montgomery (Jaeger et al., 2014a); (iii) replacing the clinical keyword lexicon with a frozen medical language model as a step-level verifier (Fan et al., 2025); (iv) per-step spatial evaluation rubrics adapted from ChestX-Reasoner’s RadRScore; (v) ablations over the five reward component weights in Table 1; and (vi) extension to multi-lesion localization.

The broader message of TB-ViTAR is that the design space for medical VLM training goes well beyond the accuracy–efficiency trade-off that dominates current literature. How a model reasons—whether its intermediate steps are spatially accurate, clinically specific, and anatomically coherent—matters as much as whether it reaches the correct final answer.

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